

TriNetra: Real-Time RGB-T Threat Intelligence

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Abstract—RGB-thermal (RGB-T) perception combines complementary visible-spectrum and thermal information and is particularly attractive for surveillance under low illumination and adverse visual conditions. However, the usefulness of either modality can vary with scene quality, sensor degradation, blur, saturation, low contrast, and cross-modal misalignment. Conventional fusion mechanisms may therefore propagate unreliable information instead of suppressing it.

This paper proposes *TriNetra*, a reliability-aware RGB-T surveillance framework centered on the TriNetra Reliability Coefficient (TRC). The primary research hypothesis is that explicitly estimating the reliability of the thermal modality before RGB-T fusion can improve the robustness of multimodal object detection compared with fixed-weight fusion and single-modality processing, particularly under degraded thermal conditions. TRC combines five interpretable cues: histogram entropy-based contrast, Laplacian-variance gradient strength, saturation/clipped-pixel ratio, normalized mutual information between RGB and thermal observations, and alignment quality.

The current implementation provides RGB-T synchronization, preprocessing, alignment handling, TRC computation, validation analysis, and controlled thermal-degradation testing. On the supplied 80-frame validation subset, TRC has a mean of 0.758 with a standard deviation of 0.024 and no sample below the configured 0.30 threshold. Under controlled degradation, representative TRC values decrease from clean thermal input to blur, saturation, and flat-response conditions. These observations provide preliminary evidence consistent with the hypothesis that TRC can act as an interpretable reliability control signal.

The paper defines the proposed fusion architecture, research hypotheses, baseline comparisons, ablation strategy, and evaluation protocol. End-to-end detection improvements are treated as a hypothesis to be experimentally validated rather than as an established result.

Index Terms—RGB-T object detection, multispectral surveillance, thermal reliability, adaptive fusion, modality failure, YOLOv8, threat intelligence, multimodal perception.

I. INTRODUCTION

Reliable visual perception is a fundamental requirement for surveillance systems deployed in environments where illumination, atmospheric conditions, or sensor quality cannot be guaranteed. Conventional RGB cameras provide rich colour, texture, and contextual information, but their effectiveness can decrease substantially in darkness, glare, fog, smoke, rain, and low-contrast scenes. Thermal infrared sensors offer complementary information because their measurements are less dependent on visible illumination; however, thermal imagery generally provides less colour and texture information and can itself be affected by blur, saturation, weak thermal contrast, sensor interference, and registration errors. The complementary nature of visible and thermal sensing has consequently motivated extensive research in RGB-T image analysis, including fusion,

pedestrian detection, object detection, tracking, and related tasks [1].

Recent RGB-T object detectors increasingly use dual-stream networks and feature-level fusion. TFDet, for example, addresses noisy fused feature maps through target-aware fusion and feature refinement, demonstrating the importance of suppressing background and false-positive features [2]. Quality-aware decision fusion has also been investigated through evidential deep learning and Dempster-Shafer theory to explicitly handle uncertainty and modality failure [3]. More recent approaches such as DENet use diffusion-based feature reconstruction, redundant-information suppression, and illumination-aware weighting to mitigate modality failure [4]. EEf further investigates energy-based feature enhancement and adaptive feedback for RGB-T feature fusion [5]. GEM-YOLO demonstrates that gated multimodal fusion can improve real-time RGB-T detection while controlling computational cost [6]. These studies collectively indicate that multimodal detection performance depends not only on the availability of two sensors, but also on how their information is weighted and combined.

The central observation motivating TriNetra is that *modality availability does not imply modality reliability*. A thermal frame may be present but provide poor information for a particular scene. Similarly, a visible frame may be severely degraded while its paired thermal image remains useful. A fusion mechanism that treats both streams identically can therefore introduce unreliable information into the fused representation. This issue is closely related to the broader multimodal fusion problem, in which complementary, redundant, and unreliable information must be integrated without allowing one source to dominate incorrectly [7].

TriNetra addresses this problem by introducing an explicit, interpretable thermal reliability signal before fusion. The proposed TriNetra Reliability Coefficient (TRC) aggregates five low-cost image-quality and cross-modal cues into a scalar in $[0, 1]$. The score is subsequently used to determine the relative contribution of the thermal and RGB streams. The current implementation uses a scalar frame-level TRC, while a spatial reliability map is retained as an extension for future region-adaptive fusion. The project implementation documents specify a pipeline consisting of RGB-T synchronization, preprocessing, TRC computation, TRC-gated fusion, YOLO-based detection, tracking, behaviour analysis, threat reasoning, and explainable alert generation.

The contribution of this work is therefore not a claim that RGB-T fusion itself is novel, but a surveillance-oriented

framework that places an interpretable, image-quality-driven reliability estimate between preprocessing and cross-modal fusion. The principal contributions are:

- 1) An interpretable thermal reliability coefficient, TRC, computed from five normalized quality cues.
- 2) A reliability-aware RGB-T fusion formulation in which the thermal contribution is controlled by TRC.
- 3) A reproducible stress-test protocol for evaluating whether TRC decreases under controlled thermal degradation.
- 4) A modular surveillance architecture connecting multimodal perception with detection, tracking, threat reasoning, and explainable alerts.
- 5) A controlled ablation protocol comparing RGB-only, thermal-only, fixed RGB-T, TRC-gated RGB-T, and future learned-weight variants.

A. Research Hypotheses

The primary hypothesis investigated in this work is:

H1: Reliability-aware RGB-T fusion using the TriNetra Reliability Coefficient will provide more robust object detection than single-modality processing and fixed-weight RGB-T fusion, particularly when one modality is degraded.

A secondary hypothesis is:

H2: TRC will decrease systematically as thermal-image quality degrades, making it suitable as an interpretable control signal for adaptive multimodal fusion.

The current validation and stress-test experiments provide preliminary evidence relevant to H2. H1 is intentionally retained as a testable research hypothesis because final end-to-end detection results are not reported in this version.

The paper deliberately distinguishes the *implemented and experimentally verified* TRC pipeline from downstream modules that remain subject to final end-to-end validation. This distinction is necessary to avoid reporting unverified detection or threat-assessment performance.

II. RELATED WORK

A. RGB-T Image Analysis

RGB-T analysis exploits complementary visible and thermal measurements and has been investigated extensively for detection, segmentation, fusion, tracking, and surveillance applications [1]. The broad literature includes image fusion, salient object detection, semantic segmentation, pedestrian detection, tracking, and person re-identification [1]. From a system perspective, multimodal fusion can be viewed at different stages of the processing pipeline, including input-level, feature-level, and decision-level integration. Modern multimodal learning further introduces attention, encoder-decoder, graph, generative, and constraint-based mechanisms [7].

For surveillance applications, the main advantage of RGB-T sensing is that the modalities exhibit different failure characteristics. RGB is typically rich in texture and colour but

strongly dependent on visible illumination. Thermal sensing can preserve object signatures when illumination is poor, but may lose fine texture or become unreliable under sensor-specific degradations. This complementarity makes reliability-aware fusion an important design consideration.

B. Adaptive and Target-Aware Fusion

TFDet proposes a target-aware fusion strategy for RGB-T pedestrian detection. Its fusion-refinement design examines parallel and cross-channel similarities and uses a segmentation branch to separate pedestrian-related features from background features [2]. The reported results on KAIST, LLVIP, FLIR, and M3FD demonstrate the value of suppressing irrelevant fused features.

GEM-YOLO introduces an adaptive multimodal gated fusion mechanism together with lightweight architectural components for real-time RGB-T detection [6]. The work is especially relevant to TriNetra because both approaches recognize that fixed or insufficiently adaptive fusion can be problematic. TriNetra differs in that the gating signal is explicitly tied to a hand-designed, interpretable thermal reliability score rather than being learned entirely inside the detector.

EEF introduces energy-score-guided feature enhancement, efficient cross-modal fusion, and adaptive feedback to improve modality representation [5]. This provides another example of using an explicit signal to influence multimodal feature processing. TriNetra instead emphasizes a low-cost quality signal that can be inspected independently of detector confidence.

C. Modality Failure and Quality-Aware Decision Making

Wang *et al.* specifically investigate RGB-T detection under failure scenarios and propose DENet, which combines diffusion-based information reconstruction, redundant-information suppression, and a kernel-similarity-aware illumination module for dynamic modality weighting [4]. Their work directly motivates the need to evaluate multimodal systems under nonroutine conditions rather than only on clean benchmark data.

Yu *et al.* propose quality-aware decision fusion using evidential deep learning and Dempster-Shafer theory [3]. Their approach expresses uncertainty at the decision level and reports improved performance under sensor interference and failure conditions. TriNetra shares the quality-awareness objective but operates earlier in the pipeline, estimating thermal reliability before feature fusion.

D. Research Gap

The supplied literature suggests three recurring directions: adaptive fusion, explicit treatment of modality failure, and robust decision making. However, the TriNetra design targets a different combination of properties. It seeks a *simple, interpretable, detector-independent reliability signal* that can be computed before fusion and used to modulate the thermal contribution. The intended framework therefore provides:

- an explicit numerical reliability signal;
- interpretable components that can be inspected individually;
- a direct connection between reliability and fusion weights;

- a degradation stress test independent of detector training;
- a modular interface that can be connected to different detectors.

This positioning is complementary to learned attention and decision-fusion approaches rather than a replacement for them.

III. TRINETRA SYSTEM ARCHITECTURE

The TriNetra framework is organized as a modular pipeline. The project documentation describes the intended sequence as dual-sensor acquisition, synchronization and preprocessing, TRC computation, reliability-aware fusion, YOLO detection, tracking and behaviour analysis, threat reasoning, and explainable alert generation. The current as-built implementation has completed the project scaffold, paired LLVIP data pipeline, and the alignment, denoising, normalization, and TRC stages. TRC-gated YOLOv8 fusion is the next major experimental stage.

A. Design Principle

Let I_R denote the visible image and I_T the paired thermal image. The central design principle is:

$$R_T = f(I_R, I_T) \in [0, 1], \quad (1)$$

where R_T is the thermal reliability estimate. A high value indicates that the thermal stream contains useful information for the current paired observation, while a low value indicates that its contribution should be reduced.

The implementation documentation specifies that LLVIP is treated as pre-aligned and that a pass-through identity transformation is used for the dataset. For general cases, homography estimation can use ORB/SIFT feature matching and RANSAC.

IV. TRINETRA RELIABILITY COEFFICIENT

A. TRC Design

The TRC is an unsupervised scalar reliability measure in $[0, 1]$. It combines five cues:

- 1) thermal contrast/information content;
- 2) gradient or edge strength;
- 3) saturation or clipped-pixel ratio;
- 4) RGB–thermal mutual information;
- 5) spatial alignment/reprojection quality.

The supplied TRC design documents specify weighted combination after normalization:

$$\text{TRC} = \sum_{i=1}^5 w_i S_i, \quad \sum_{i=1}^5 w_i = 1, \quad w_i \geq 0, \quad (2)$$

where $S_i \in [0, 1]$ is the normalized score for cue i .

B. Contrast Score

Histogram entropy is used to quantify information content:

$$H = - \sum_{j=0}^{L-1} p_j \log_2 p_j, \quad (3)$$

where p_j is the normalized probability of intensity bin j . The entropy is normalized to obtain:

$$S_C = \text{norm}(H) \in [0, 1]. \quad (4)$$

Higher entropy generally corresponds to richer intensity variation, although entropy alone does not guarantee semantic usefulness.

C. Gradient Score

The gradient cue uses Laplacian variance as a sharpness indicator:

$$S_G = \frac{\sigma_{\nabla^2 I_T}^2}{\sigma_{\nabla^2 I_T}^2 + K}, \quad (5)$$

where $\sigma_{\nabla^2 I_T}^2$ denotes the variance of the Laplacian response and K is a stabilizing constant. The resulting score is normalized to $[0, 1]$.

D. Saturation Score

Thermal images containing excessive clipped pixels may lose useful information. Let N be the total number of pixels and N_c the number of pixels classified as clipped. The clipped-pixel fraction is:

$$r_c = \frac{N_c}{N}. \quad (6)$$

A reliability-oriented score can be written as:

$$S_S = 1 - r_c, \quad (7)$$

after any implementation-specific normalization and clipping.

E. Cross-Modal Mutual Information

Mutual information is used to measure statistical dependence between paired RGB and thermal observations. The implementation design uses normalized mutual information:

$$S_{MI} = \text{NMI}(I_R, I_T). \quad (8)$$

A higher value indicates stronger cross-modal statistical correspondence, subject to the limitations of mutual-information-based similarity measures.

F. Alignment Quality

Spatial misalignment can reduce the usefulness of feature fusion. If d denotes a mean reprojection error, the design document specifies an exponential reliability mapping:

$$S_A = \exp\left(-\frac{d}{\lambda}\right), \quad (9)$$

where λ controls the decay rate. Thus, larger alignment error produces a lower reliability score.

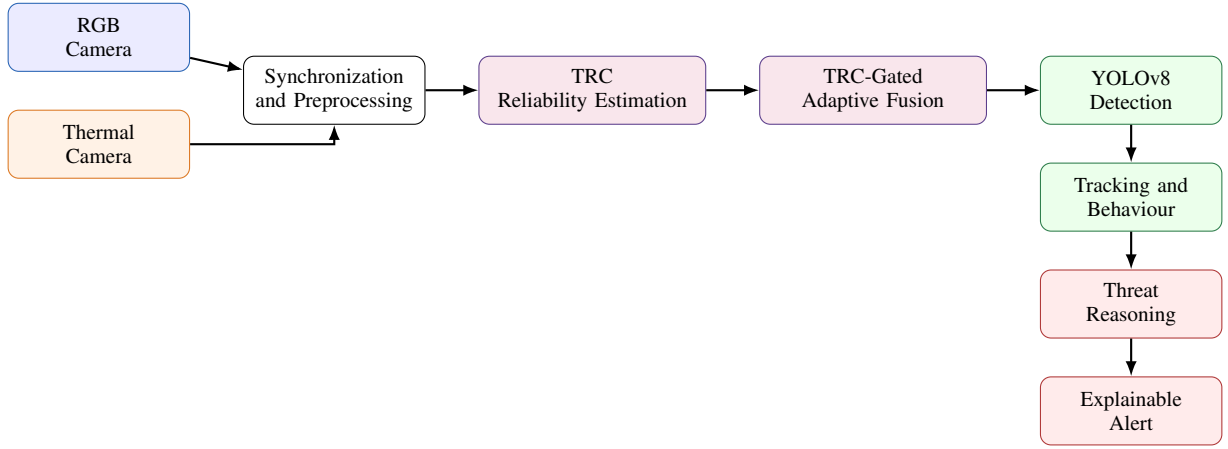


Fig. 1: TriNetra end-to-end surveillance architecture. The solid blocks from sensing through TRC represent the implemented Phase 1–3 pipeline; downstream fusion, detection, tracking, threat reasoning, and alert modules form the intended end-to-end research pipeline and must be supported by final experiments before performance claims are made.

G. TRC-to-Fusion Mapping

For the frame-level formulation, the thermal and RGB weights are defined as complementary quantities:

$$W_T = \text{TRC}, \quad (10)$$

$$W_R = 1 - \text{TRC}, \quad (11)$$

which gives:

$$W_T + W_R = 1. \quad (12)$$

The corresponding feature-level formulation is:

$$F_{\text{fused}} = W_R F_R + W_T F_T, \quad (13)$$

where F_R and F_T are RGB and thermal feature representations.

The implementation plan additionally proposes a residual formulation at YOLO feature levels:

$$F_{\text{out}} = F_R + \mathcal{B}(\text{TRC} \cdot F_T), \quad (14)$$

where $\mathcal{B}(\cdot)$ is a learnable thermal fusion block. This residual version should be treated as the planned Phase-4 implementation unless the final code confirms otherwise.

V. TRC COMPUTATION PROCEDURE

The as-built preprocessing pipeline is intentionally mild because aggressive denoising can remove small pedestrian edges. The implementation documents describe bilateral/NLM-style denoising, CLAHE or percentile-based normalization, and dataset-aware alignment handling.

Algorithm 1 Thermal Reliability Coefficient Computation

Require: RGB frame I_R , thermal frame I_T

Ensure: $\text{TRC} \in [0, 1]$

- 1: Align I_R and I_T
- 2: Denoise thermal frame while preserving edges
- 3: Normalize thermal intensity
- 4: Compute contrast score S_C
- 5: Compute gradient score S_G
- 6: Compute saturation score S_S
- 7: Compute cross-modal mutual-information score S_{MI}
- 8: Compute alignment score S_A
- 9: Normalize all five scores to $[0, 1]$
- 10: Compute $\text{TRC} = \sum_{i=1}^5 w_i S_i$
- 11: Clip TRC to $[0, 1]$
- 12: **return** TRC

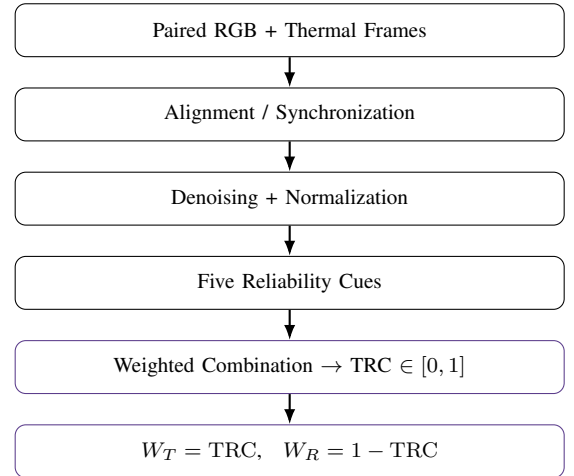


Fig. 2: TRC computation and reliability-to-fusion mapping.

VI. IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Development Stack

The supplied implementation documentation identifies Python and C++ as programming languages, PyTorch and

TABLE I: TriNetra implementation stack

Layer	Tools / Components
Programming	Python, C++
Deep learning	PyTorch; Ultralytics YOLOv8/YOLO family
Computer vision	OpenCV, NumPy, Pillow, scikit-image
Augmentation	Albumentations / synchronized RGB-T transforms
Data format	COCO conversion and paired RGB-T DataLoader
Reliability	Entropy, Laplacian variance, clipping, NMI, reprojection error
Tracking	ByteTrack / DeepSORT-type modular tracker
Visualization	Matplotlib, Plotly, TensorBoard
Deployment	NVIDIA CUDA GPU; planned API and ONNX Runtime path
Development	VS Code, Git, GitHub; configuration-driven YAML pipeline

TABLE II: Datasets identified for TriNetra evaluation

Dataset	Role in TriNetra
LLVIP	Primary paired RGB-T low-light surveillance dataset; current TRC pipeline verified end-to-end on LLVIP.
FLIR	Supplementary thermal/RGB-T evaluation source for robustness and generalization experiments.
Additional RGB-T benchmarks	Optional cross-dataset validation under varied day/night and degradation conditions.

Ultralytics YOLO as the deep-learning stack, OpenCV and NumPy for computer vision, Matplotlib/Plotly for visualization, and CUDA-enabled NVIDIA GPU acceleration for deployment-oriented experiments. The broader planned stack includes Albumentations, Pillow, scikit-image, TorchMetrics, COCO tools, TensorBoard, and optional FastAPI/Uvicorn and ONNX Runtime components.

B. Dataset and Data Preparation

LLVIP is the principal paired visible-thermal dataset in the current implementation. The supplied project documentation describes Phase 2 as complete, including paired RGB-T loading, COCO conversion, and synchronized augmentation. FLIR thermal data and additional RGB-T benchmark data are identified as supplementary evaluation sources.

C. Augmentation

RGB and thermal images must undergo synchronized geometric transformations so that corresponding pixels remain spatially meaningful. The augmentation pipeline should preserve pair identity and apply equivalent geometric transforms to both modalities. Photometric transformations should be modality-aware because visible intensity and thermal intensity have different physical meanings.

D. Detection Configuration

The research plan specifies a YOLOv8-based detector with two modality streams exposing multi-scale features at P3/P4/P5. The visible branch can be initialized from pretrained weights, while the thermal branch and fusion blocks are trained using transfer learning. The recommended strategy is to initially freeze the pretrained visible backbone/neck/head, train the newly introduced thermal and fusion components, and optionally perform low-learning-rate fine-tuning.

TABLE III: Required ablation and baseline experiments

ID	Configuration
B1	RGB-only YOLO detector
B2	Thermal-only YOLO detector
B3	Conventional RGB-T fusion with fixed/equal weighting
B4	TRC-gated RGB-T fusion (TriNetra-v1)
B5	TRC + learned adaptive weighting using a small MLP (TriNetra-v2, optional research extension)

The exact YOLOv8 variant, image size, batch size, number of epochs, learning rate, optimizer, hardware, and confidence/IoU thresholds must be filled from the final training logs rather than inferred from the project documents.

VII. EXPERIMENTAL DESIGN

The experimental design is organized to isolate the effect of reliability-aware fusion.

A. Baseline Comparisons

The following models should be evaluated under identical data splits and training/evaluation conditions:

This design follows the implementation strategy documented for TriNetra: establish a fixed-weight baseline first, then replace the fixed weighting mechanism with a learned MLP so that the effect of adaptive weighting can be isolated.

B. Evaluation Metrics

Detection performance should be reported using:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (15)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (16)$$

and

$$F_1 = \frac{2PR}{P + R}. \quad (17)$$

Mean average precision should be reported at IoU = 0.50 and, where supported by the evaluation framework, across IoU thresholds from 0.50 to 0.95. Runtime should be measured using frames per second (FPS) and end-to-end latency. The same hardware and measurement procedure must be used across baselines.

C. Reliability Stress Testing

The TRC module should be evaluated independently of detector performance. The current stress-test conditions are:

- 1) clean thermal input;
- 2) blurred thermal input;
- 3) saturated/clipped thermal input;
- 4) flat or low-contrast thermal input;
- 5) controlled RGB–thermal alignment perturbation.

The objective is to test whether TRC changes monotonically or consistently in the expected direction as thermal information becomes less reliable.

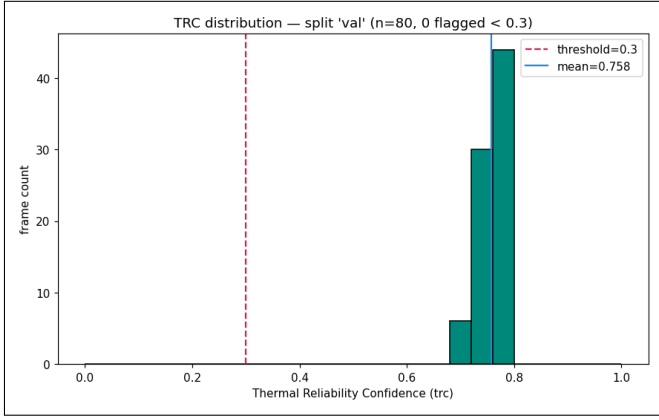


Fig. 3: TRC distribution over the validation split. The supplied evaluation output reports $n = 80$, mean TRC = 0.758, and zero frames below the 0.30 threshold.

TABLE IV: Observed TRC under thermal degradation

Condition	TRC
Clean	0.67
Blur	0.62
Saturation	0.38
Flat / low-contrast	0.40

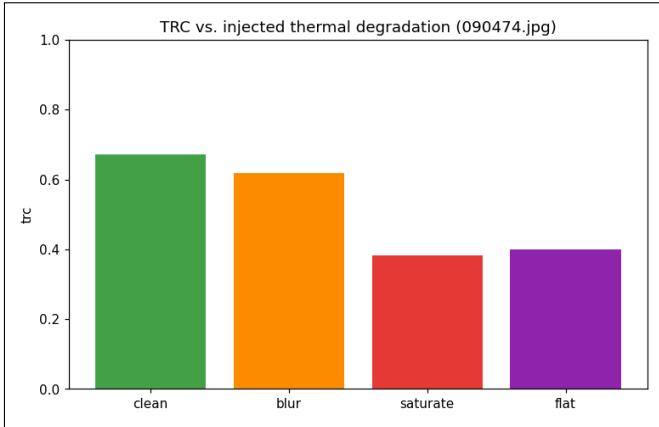


Fig. 4: TRC response under the supplied thermal degradation stress test.

VIII. RESULTS

A. TRC Distribution on Validation Data

The current validation output contains 80 frames. The mean thermal reliability confidence is 0.758 and no frame falls below the configured 0.30 flagging threshold.

The concentration of scores around the higher-reliability region indicates that the current validation subset is generally assigned a usable thermal confidence. This observation should not be interpreted as proof of detection improvement; it validates only the empirical distribution of the reliability estimator.

B. TRC Under Synthetic Thermal Degradation

The controlled stress test produces:



Fig. 5: Representative thermal preprocessing and TRC-map outputs.

The results show a substantial reduction in TRC for saturation and flat thermal responses relative to the clean condition. Blur also produces a decrease, although the reduction is smaller. This supports the intended interpretation that the reliability score is sensitive to the tested degradation modes.

C. Per-Frame TRC Component Examples

Two supplied preprocessing reports provide frame-level examples. For frame 030396.jpg, the reported TRC is 0.615 with component values of contrast 0.92, gradient 0.11, saturation 0.99, mutual information 0.05, and alignment error reported as 1.00. For frame 091028.jpg, the corresponding TRC is 0.640 with contrast 0.95, gradient 0.20, saturation 0.98, mutual information 0.07, and alignment error reported as 1.00.

The repeated alignment-error value of 1.00 requires verification before a final journal/conference version is submitted. In particular, the implementation should confirm whether this quantity denotes normalized error, normalized quality, or a metric that has saturated at its upper bound. The paper should use the exact implementation convention.

D. Qualitative RGB-T Observations

The supplied paired visualizations show complementary information across RGB and thermal modalities. In low-light scenes, RGB retains scene texture and contextual structure while thermal emphasizes warm objects and pedestrians. The overlay visualizations illustrate the potential value of combining the two streams.

E. Detection Performance

The final detector results must be populated after the controlled YOLOv8 training and evaluation runs. The following table is intentionally left with placeholders.

No numerical superiority claim should be made until this table is filled from the final experiments.

IX. ABLATION AND SENSITIVITY ANALYSIS

A. Five-Cue TRC Ablation

To establish that the TRC is not dominated by a single cue, each component should be removed individually while all remaining conditions are kept constant.

B. Degradation-Severity Sensitivity

For each degradation type, multiple severity levels should be generated. Let δ denote degradation severity. A useful diagnostic is:

$$\rho_{\text{TRC}} = \text{corr}(\text{TRC}, -\delta), \quad (18)$$

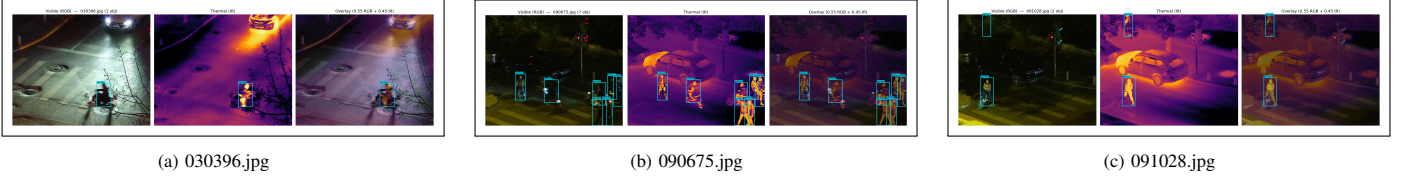


Fig. 6: Representative RGB-T paired observations and detections from the validation examples supplied with the current implementation.

TABLE V: End-to-end detection comparison. Replace all “–” entries with values from the same test protocol.

Method	Precision	Recall	F1	mAP@50	mAP@50:95	FPS	Notes
RGB-only	–	–	–	–	–	–	Baseline
Thermal-only	–	–	–	–	–	–	Baseline
Fixed RGB-T	–	–	–	–	–	–	Fixed weighting
TriNetra-v1	–	–	–	–	–	–	TRC-gated fusion
TriNetra-v2	–	–	–	–	–	–	TRC + MLP weights

TABLE VI: TRC cue ablation template

Configuration	S_C	S_G	S_S	S_{MI}	S_A	TRC
All five cues	✓	✓	✓	✓	✓	–
No contrast	–	✓	✓	✓	✓	–
No gradient	✓	–	✓	✓	✓	–
No saturation	✓	✓	–	✓	✓	–
No mutual information	✓	✓	✓	–	✓	–
No alignment	✓	✓	✓	✓	–	–

where a positive correlation indicates that TRC decreases as degradation increases when the sign convention is defined accordingly. The final paper should report the correlation coefficient and confidence interval only after repeated experiments.

C. Fusion-Weight Sensitivity

Because $W_T = \text{TRC}$ and $W_R = 1 - \text{TRC}$, the effective thermal contribution should be inspected across the validation distribution. A histogram of W_T is equivalent to the TRC distribution under the current mapping. This provides an interpretable link between reliability estimation and fusion behaviour.

X. SURVEILLANCE AND THREAT-INTELLIGENCE LAYER

TriNetra is intended to extend beyond object detection toward surveillance-oriented temporal reasoning. The architecture includes multi-object tracking, behaviour analysis, contextual threat assessment, and explainable alert generation. These components are modular and should not be conflated with the current TRC validation results.

A. Tracking

The detector produces object bounding boxes and confidence scores. A multi-object tracker associates detections across consecutive frames and maintains persistent object identities. Candidate tracking outputs include object IDs, trajectories, dwell time, motion direction, and estimated speed.

B. Behaviour Analysis

Behaviour analysis can use temporal features such as:

$$v_t = \frac{\|\mathbf{p}_t - \mathbf{p}_{t-\Delta t}\|}{\Delta t}, \quad (19)$$

where $\mathbf{p}_t$ is the object centroid at time t . Directional change, dwell time, and trajectory deviation can be combined with spatial context.

C. Contextual Threat Score

The copyright-level design describes hierarchical threat reasoning using contextual cues such as zone violation, abnormal movement, density/congestion, and risk factors. A generic formulation is:

$$T = \alpha Z + \beta B + \gamma M + \delta D + \eta C, \quad (20)$$

where Z is zone violation, B behaviour abnormality, M motion-pattern evidence, D density/congestion, and C contextual confidence. The coefficients should be calibrated empirically and must not be presented as validated values unless measured.

D. Explainability

For each alert, the system is intended to expose the principal evidence that contributed to the decision, such as:

- detected object and class;
- object ID and trajectory;
- RGB and thermal reliability;
- contextual event;
- threat score;
- final threat level;
- timestamp and location/zone identifier.

This design provides a human-readable chain from sensing to alert rather than producing an unexplained binary output.

XI. COMPARISON WITH RELATED RGB-T STRATEGIES

This comparison is conceptual and describes the architectural emphasis of each approach; it is not a claim that the methods use identical datasets, detector backbones, or experimental protocols.

TABLE VII: Conceptual comparison of TriNetra with supplied related RGB-T approaches

Criterion	TFDet	QDFM	DENet	GEM-YOLO	TriNetra
Primary focus	Target-aware fusion/refinement	Quality-aware decision fusion	Failure-aware feature re-construction	Lightweight gated fusion	Reliability-aware surveillance pipeline
Reliability signal	Learned feature relationships	Evidential uncertainty	Illumination/similarity-aware weighting	Learned multimodal gate	Explicit five-cue TRC
Fusion stage	Feature fusion	Decision fusion	Feature processing/fusion	Feature fusion	Pre-fusion reliability gating + feature fusion
Modality failure	Indirectly handled through refinement	Explicitly targeted	Explicitly targeted	Adaptive gating	Explicit stress-tested reliability score
Interpretability of reliability	Limited	Evidence-based	Learned	Learned gate	High; individual cues exposed
Detector dependence	Detection-oriented	Detection-oriented	Detection-oriented	YOLO-oriented	Designed as detector-agnostic module
Tracking/threat layer	No central focus	No central focus	No central focus	No central focus	Included in system architecture
Real-time orientation	Reported competitive efficiency	Not primary contribution	Not primary contribution	Strong emphasis	Deployment-oriented design

XII. DISCUSSION

A. Interpretation of Current Findings

The current results provide preliminary evidence relevant to H2: whether an interpretable thermal reliability score can be computed consistently over paired RGB-T observations and whether it responds to controlled thermal degradation. The validation distribution demonstrates that the score is numerically stable within the tested subset, while the stress test demonstrates lower values for several degradation modes.

The stress-test result is particularly important because it is independent of detector training. It tests the reliability estimator itself. However, a detector-level conclusion requires the next experiment: the same degraded inputs must be evaluated using RGB-only, thermal-only, fixed-fusion, and TRC-gated models. Only then can a relationship between reliability-aware fusion and detection performance be established.

B. Interpretability

A principal advantage of the current formulation is that a TRC value can be decomposed into its five constituent cues. This allows an engineer to inspect whether a low score is caused by poor contrast, weak edge information, excessive clipping, low cross-modal agreement, or poor alignment. This is more transparent than a learned fusion gate whose internal weighting is not directly interpretable.

C. Limitations

Several limitations must be addressed in the final version.

First, the current TRC implementation is frame-level rather than fully spatial. A single score can suppress or enhance the thermal stream globally even when reliability varies strongly within a frame. A spatial TRC map is therefore a natural extension.

Second, the five cues are hand-designed. Their relative weights may not be optimal across datasets, sensors, or environments. The planned MLP-based adaptive weighting experiment provides a way to investigate learned weighting while preserving the same experimental pipeline.

Third, mutual information and alignment measures can be sensitive to preprocessing and registration. The repeated alignment-related value reported as 1.00 in the supplied per-frame outputs must be inspected before publication.

Fourth, the current paper distinguishes the verified TRC stage from the planned end-to-end detection and threat modules. The latter require controlled experiments before quantitative claims can be made.

Finally, synthetic degradation does not perfectly reproduce real sensor failure. Future evaluation should include naturally occurring adverse conditions and, where possible, real sensor interference.

XIII. FUTURE WORK

The immediate research objective is to complete the TRC-gated YOLOv8 fusion implementation and conduct the baseline/ablation experiments. The next stage is the learned weighting variant in which a small MLP receives the five quality cues and predicts adaptive modality weights. This provides a clean comparison between interpretable fixed weighting and learned adaptive weighting.

A second direction is region-adaptive reliability. Instead of a scalar TRC, the system can estimate:

$$\mathbf{R}_T(x, y) \in [0, 1], \quad (21)$$

allowing different regions of a frame to contribute thermal information at different strengths.

Cross-modal attention and lightweight edge deployment are additional directions identified in the project documentation. Wider benchmarking across LLVIP, FLIR, and other RGB-T datasets is also necessary to determine whether the reliability formulation generalizes beyond the current validation subset.

The threat-intelligence layer can subsequently be evaluated using tracking metrics, event-detection precision/recall, alert latency, and false-alarm rate. Such evaluation should be conducted separately from object-detection metrics.

XIV. CONCLUSION

This paper presented TriNetra, a reliability-aware RGB-T surveillance framework centred on the TriNetra Reliability Coefficient. The approach explicitly estimates thermal reliability from five normalized cues—contrast, gradient strength, saturation, cross-modal mutual information, and alignment quality—and maps the resulting score to complementary modality weights for adaptive fusion.

The current implementation provides an end-to-end TRC preprocessing and evaluation pipeline. On the supplied validation output, 80 frames produced a mean TRC of 0.758 with no frame below the 0.30 flagging threshold. In a controlled degradation test, TRC decreased from 0.67 for clean thermal input to 0.62 under blur, 0.38 under saturation, and 0.40 for flat/low-contrast input. These findings support the intended sensitivity of the reliability estimator to the tested thermal degradation modes.

The central remaining experimental question is whether this reliability signal improves object detection compared with single-modality and fixed-fusion baselines. The paper therefore defines a controlled YOLOv8 evaluation and ablation protocol, including RGB-only, thermal-only, fixed RGB-T, TRC-gated fusion, and optional learned-weight variants. Once those experiments are completed, the placeholder quantitative results can be replaced with measured values and statistical analysis.

TriNetra is consequently positioned as a modular bridge between low-level multimodal quality assessment and higher-level surveillance intelligence. Its reliability signal is intended to provide a transparent control mechanism for multimodal fusion while allowing the detector, tracker, and downstream reasoning components to remain modular.

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TABLE VIII: Resolved TriNetra implementation parameters

Parameter	Value
Dataset	LLVIP
Total RGB-T pairs	15,488
Total annotations	42,432
Train / Validation / Test	9,903 / 2,122 / 3,463
Native resolution	1280 × 1024
Model input resolution	640 × 512
RGB channels	3
Thermal channels	1
Detector	YOLOv8m
Pretrained weights	COCO pretrained
Ultralytics version	8.4.89
Fusion scales	P3 / P4 / P5
Fusion strides	8 / 16 / 32
TRC cue weights	0.2 each
Gradient normalization constant K	15
TRC type	Scalar frame-level
Spatial TRC map	Implemented as optional extension
LLVIP alignment	Identity / pass-through
Alignment reprojection error	0 for aligned LLVIP pairs
Validation frames	80
Mean TRC	0.758
TRC standard deviation	0.024
Minimum TRC	0.694
Maximum TRC	0.787
Frames below TRC 0.30	0 / 80

APPENDIX A

RESOLVED IMPLEMENTATION CONFIGURATION

The following parameters are fixed according to the current TriNetra implementation and project configuration.

APPENDIX B

HYPOTHESIS-STAGE SCOPE

This manuscript is intentionally presented at the hypothesis and prototype-validation stage. The measured TRC validation and thermal degradation results are reported as preliminary evidence relevant to H2. The principal H1 claim—improved end-to-end RGB-T object detection through reliability-aware fusion—is formulated as a testable hypothesis and is not presented as a confirmed result.

No unmeasured mAP, precision, recall, F1, FPS, latency, tracking accuracy, or threat-classification accuracy is claimed in this version.